MACHINE LEARNING · PERSONAL PROJECT

Workout Calorie Analyzer

Predicts calories burned in a workout and groups workout routines by intensity, using wearable fitness data.

Built using a practice dataset · Kasthuri Sathish

The problem

Wearables collect a lot of workout data, but people rarely get a clear answer to two questions: how many calories a session really burned, and what kind of workout routine they follow.

What I built

A machine learning pipeline that predicts calorie burn from workout data, and a clustering model that groups sessions into routines based on heart rate zones and duration.

How it works

Wearable workout data → Clean data and treat outliers (IQR) → Feature engineering → Linear Regression and Random Forest → K-Means routine groups → Calorie prediction and insights

Key product decisions

- Combined prediction and clustering, so users get both a number and a picture of their routines.
- Compared a simple model with a stronger one, to see whether the extra complexity paid off.
- Treated outliers with IQR, so faulty sensor readings didn't distort the model.
- Used the Elbow Method and Silhouette analysis to choose clusters that separate light, moderate, and intense routines.

Testing and learnings

- Random Forest outperformed Linear Regression with a higher R^2, capturing non-linear links between heart rate and duration.
- Scaled features with StandardScaler so large raw values, like step counts, didn't bias the K-Means clustering.
- Duration and heart rate tend to drive calorie burn more than other features.
- Limitation: trained on a practice dataset, so results may not hold for every person.

Tech stack

Python · Scikit-Learn · Pandas · Supervised Regression · K-Means Clustering · Feature Engineering